Non-Stationary Gaussian Processes and Spectral Methods

A Complete Mathematical Reference

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1. What Stationarity Means and Why It Fails

A kernel function $k(x, x′)$ is **stationary** if it depends only on the difference $\tau = x - x′$ rather than on the absolute positions x and $x′$ individually:

$$k(x, x′) = k(x - x′) = k(\tau)$$

This is a translation invariance assumption. It says the function has the same statistical properties everywhere in the input space: the same smoothness, the same variance, the same length scale. A Matérn or squared exponential kernel is stationary by construction — $k(x, x′) = k(|x - x′|)$ depends only on the distance between inputs.

For many physical and real-world processes, stationarity is a poor assumption. Consider reward as a function of intensity. Near zero intensity, actions are so weak that the reward function is likely flat and uninformative — slowly varying, well-described by a long length scale. Near maximum intensity, small changes may cross important thresholds and produce dramatically different outcomes — rapidly varying, better described by a short length scale. A stationary kernel imposes the same length scale across this entire range, which misrepresents the function's structure.

Non-stationary processes arise naturally in many domains: spatial statistics (soil properties vary smoothly in flat regions but rapidly near geological features), time-series modeling (financial volatility clusters and shifts), and reinforcement learning (reward landscapes change as the environment adapts).

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2. Bochner's Theorem and the Spectral Representation

A deep result connecting stationary kernels to Fourier analysis provides both theoretical insight and practical algorithms.

Bochner's Theorem (1933): A continuous, bounded, complex-valued function $k(\tau)$ is a positive definite (valid stationary kernel) function if and only if it is the Fourier transform of a finite non-negative measure $S(\omega)$:

$$k(\tau) = \int S(\omega), e^{i\omega\tau}, d\omega$$

The measure $S(\omega)$ is called the **spectral density** of the kernel. This is a bijection between the class of valid stationary kernels and the class of non-negative spectral densities — every valid stationary kernel has a spectral density, and every non-negative spectral density defines a valid stationary kernel.

This has a beautiful interpretation. The kernel $k(\tau)$ encodes beliefs about correlation as a function of distance. The spectral density $S(\omega)$ encodes beliefs about energy at each frequency — how much the function oscillates at each scale. A kernel that concentrates S around low frequencies corresponds to smooth, slowly-varying functions. One that spreads S broadly corresponds to rough, rapidly-varying functions.

2.1 Spectral Densities of Common Kernels

For the **squared exponential kernel** $k(\tau) = \exp(-\tau^2/2\ell^2)$, the spectral density is Gaussian: $$S(\omega) = \sqrt{2\pi},\ell \cdot \exp!\left(-\frac{\ell^2 \omega^2}{2}\right)$$

For the **Matérn 5/2 kernel**, the spectral density is a Student-t distribution in frequency space: $$S(\omega) \propto \left(\frac{5}{\ell^2} + \omega^2\right)^{-3}$$

The heavier tails of the Matérn spectral density compared to the Gaussian correspond to the presence of more high-frequency content in Matérn sample functions — exactly the property that makes Matérn functions rougher than squared exponential functions.

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3. The Spectral Mixture Kernel

The spectral representation suggests a natural strategy for flexible kernel construction: instead of starting from a kernel and computing its spectral density, directly parametrize $S(\omega)$ as a flexible family and let the data determine its shape.

The **Spectral Mixture kernel** (Wilson and Adams, 2013) models $S(\omega)$ as a mixture of Gaussians:

$$S(\omega) = \sum_{q=1}^{Q} w_q, \mathcal{N}(\omega;, \mu_q, \sigma_q^2)$$

Taking the inverse Fourier transform of this mixture yields the corresponding kernel:

$$k_{\text{SM}}(\tau) = \sum_{q=1}^{Q} w_q \exp!\left(-2\pi^2 \tau^2 \sigma_q^2\right)\cos(2\pi\tau\mu_q)$$

Each component contributes a term with characteristic frequency μ_q, characteristic length scale $1/\sigma_q$, and weight w_q. This kernel can automatically discover and represent periodic structure (components near nonzero μ_q), multi-scale structure (multiple components at different σ_q), and long-range correlations (components with small σ_q).

The hyperparameters $\{w_q, \mu_q, \sigma_q\}_{q=1}^Q$ are learned by maximizing the marginal likelihood, exactly like the length scale of a standard kernel. The number of components Q is a model selection choice that controls expressiveness versus overfitting risk.

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4. Random Fourier Features

Bochner's theorem provides a computationally powerful algorithm for approximating stationary kernels. If $S(\omega)$ is normalized to be a proper probability distribution, we can write:

$$k(\tau) = \mathbb{E}_{\omega \sim S}\!\left[e^{i\omega\tau}\right] = \mathbb{E}_{\omega \sim S}\!\left[\cos(\omega\tau)\right]$$

(using only the real part for real-valued inputs). This suggests a Monte Carlo approximation: draw D random frequencies $\omega_1, \ldots, \omega_D$ from $S(\omega)$, and approximate the kernel by:

$$k(x, x') \approx \hat{k}(x, x') = \frac{1}{D}\sum_{d=1}^D \cos(\omega_d(x - x'))$$

This can be written as a dot product $\hat{k}(x, x') = \boldsymbol{\phi}(x)^T \boldsymbol{\phi}(x')$ where:

$$\boldsymbol{\phi}(x) = \frac{1}{\sqrt{D}}\left[\cos(\omega_1 x + b_1),\ \ldots,\ \cos(\omega_D x + b_D)\right]^T$$

with $b_d \sim \text{Uniform}[0, 2\pi]$. This feature map $\boldsymbol{\phi}(x)$ converts the kernel regression problem into ordinary linear regression in a D-dimensional feature space, at cost $O(nD)$ rather than $O(n^3)$.

Random Fourier Features are an important practical algorithm, but they require knowing (or estimating) the spectral density $S(\omega)$ to sample from, which means they only directly apply to kernels for which S is known in closed form.

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5. Approaches to Non-Stationary Kernels

5.1 Input Warping

The simplest approach to non-stationarity applies a monotonic transformation $g: \mathcal{X} \to \mathcal{X}$ to the input space before computing a stationary kernel:

$$k_{\text{warped}}(x, x') = k_{\text{stat}}(g(x), g(x'))$$

Where g is steep (large derivative), the effective distance between nearby inputs is large — the kernel treats them as less similar, implying faster variation. Where g is flat, the effective distance is small — nearby inputs are treated as more similar, implying slower variation.

The Kumaraswamy distribution is commonly used to parametrize g for inputs in $[0,1]$, because its two shape parameters give flexible control over the warping profile. These parameters are optimized alongside the kernel hyperparameters.

Input warping is computationally free — it adds preprocessing steps but requires no structural changes to GP inference. Its limitation is that it can only represent smoothly varying length scales, not abrupt transitions.

5.2 The Gibbs Non-Stationary Kernel

The **Gibbs kernel** (1997) allows the length scale to vary continuously across the input space as a function $\ell(x)$:

$$k_{\text{Gibbs}}(x, x') = \sqrt{\frac{2\ell(x)\ell(x')}{\ell(x)^2 + \ell(x')^2}} \cdot \exp\!\left(-\frac{(x - x')^2}{\ell(x)^2 + \ell(x')^2}\right)$$

The prefactor $\sqrt{2\ell(x)\ell(x')/(\ell(x)^2 + \ell(x')^2)}$ is the geometric-harmonic mean of the two local length scales. Crucially, this prefactor ensures the kernel remains positive definite for any choice of $\ell(x)$ — a non-trivial property that the Gibbs construction satisfies by design.

The challenge is specifying $\ell(x)$. Options range from a simple parametric function (sigmoid, polynomial) to a full GP prior over $\ell(x)$. The latter leads to **deep GPs** — hierarchical models where one GP parametrizes another — which are highly flexible but expensive to infer.

For the intensity domain $[0,1]$, a sigmoid length scale function is natural:

$$\ell(x) = \ell_{\min} + (\ell_{\max} - \ell_{\min}) \cdot \sigma(k(x - x_0))$$

where σ is the logistic function, x_0 is the transition point, k controls the steepness of the transition, and $\ell_{\min}$, $\ell_{\max}$ are the length scales at the two extremes. This four-parameter family can represent a smooth transition from one regime of smoothness to another.

5.3 Deep Kernel Learning

Deep Kernel Learning (Wilson et al., 2016) passes inputs through a neural network ϕ_w before applying a stationary kernel:

$$k_{\text{DKL}}(x, x') = k_{\text{stat}}(\phi_w(x), \phi_w(x'))$$

The network learns to warp the input space in whatever way makes the resulting kernel best fit the data. Positive definiteness is inherited from k_{stat} regardless of ϕ_w, because composing a kernel with any function preserves positive definiteness: if k is positive definite, then $k(\phi(x), \phi(x'))$ is also positive definite.

The weights w of the neural network are optimized jointly with the kernel hyperparameters by maximizing the marginal likelihood using automatic differentiation — an end-to-end training procedure that is natural in frameworks like PyTorch or TensorFlow.

Deep Kernel Learning is the most flexible approach to non-stationarity but also the most computationally demanding and most prone to overfitting with small datasets.

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6. Locally Stationary Processes and Non-Stationary Spectral Methods

The most general framework for non-stationary kernels allows the spectral density itself to vary with position:

$$k(x, x') = \int S!\left(\omega;, \frac{x + x'}{2}\right) e^{i\omega(x - x')}, d\omega$$

Here $S(\omega; u)$ is a spectral density that depends on the midpoint $u = (x + x')/2$ between the two inputs. This is called a **locally stationary** kernel: at each location u, the kernel looks approximately stationary with spectral density $S(\omega; u)$.

Recent work by Remes, Heinonen, and Kaski (2017) on non-stationary spectral mixture kernels models $S(\omega; u)$ as a mixture of Gaussians whose parameters (weights, means, variances) are themselves smooth functions of u. This gives the model the flexibility to discover different periodic structures, different smoothness levels, and different noise characteristics in different regions of the input space — all learned from data.

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7. Time-Decay Approximation for Practical Non-Stationarity

When the non-stationarity is primarily temporal — the function is changing over time rather than varying over the input space — a simple and effective approximation is to **down-**

weight older observations in the kernel matrix.

Define a time-decay weight for observation i at time t_i as:

$$w_i = \gamma^{t_{\max} - t_i}, \quad \gamma \in (0, 1)$$

where $t_{\max}$ is the most recent observation time. The weighted kernel matrix is then:

$$\tilde{K}_{ij} = \sqrt{w_i w_j} \cdot k(x_i, x_j)$$

The square-root weighting preserves positive definiteness via the congruence transformation $\tilde{K} = D K D$ where $D = \text{diag}(\sqrt{w_i})$. This is because for any vector $\mathbf{v}$: $\mathbf{v}^T (DKD) \mathbf{v} = (D\mathbf{v})^T K (D\mathbf{v}) \geq 0$ when K is positive semidefinite.

This approximation is computationally free — it requires only an elementwise multiplication of the kernel matrix — and allows the GP to adapt to slowly drifting environments without the full machinery of non-stationary kernels.

See also: Gaussian Process Regression; Multi-Output GP and Coregionalization; The Two-Timescale Bayesian Update